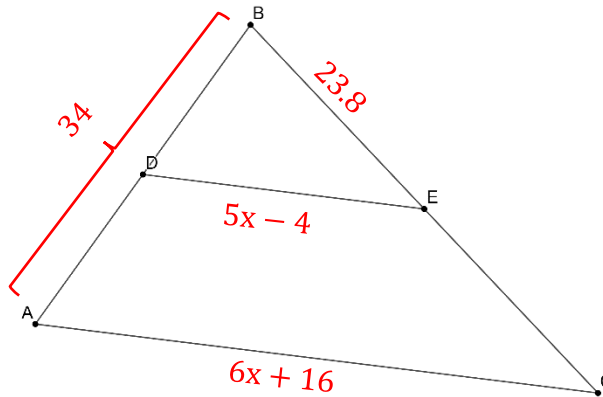


Geometry  
Unit 11  
Lesson 9 Practice

Name **Answer Key**  
Date \_\_\_\_\_  
Period \_\_\_\_\_

Use this information to answer questions 1 through 5.

Triangle  $ABC$  is shown, where  $\overline{DE}$  is the midsegment.  $AC = 6x + 16$ ,  $DE = 5x - 4$ ,  $AB = 34$ , and  $BE = 23.8$



1. Find the value of  $x$ .

$$\begin{aligned} 2(5x - 4) &= 6x + 16 \\ 10x - 8 &= 6x + 16 \\ 4x &= 24 \\ x &= 6 \end{aligned}$$

2. Identify the length of  $\overline{DE}$ .

$$DE = 5(6) - 4 = 30 - 4 = 26$$

3. Identify the length of  $\overline{AC}$ .

$$AC = 6(6) + 16 = 52$$

4. Identify the length of  $\overline{AD}$ .

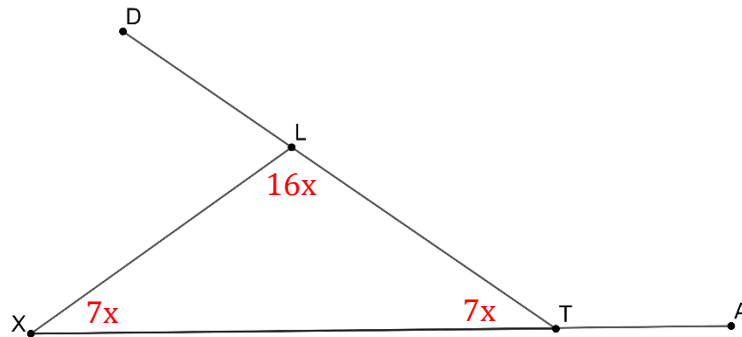
$$AD = \frac{34}{2} = 17$$

5. Calculate the perimeter of triangle  $BDE$ .

$$P = 17 + 23.8 + 26 = 66.8$$

Use this information to answer questions 6 through 10.

Triangle  $XTL$  is shown, where  $\angle XLP$  and  $\angle LTA$  are exterior angles.  $m\angle LXT = m\angle LTX = (7x)^\circ$  and  $m\angle XLT = (16x)^\circ$



6. Find the value of  $x$ .

$$\begin{aligned}7x + 7x + 16x &= 180 \\30x &= 180 \\x &= 6\end{aligned}$$

7. Find  $m\angle XLT$ .  
 $m\angle XLT = 16(6) = 96^\circ$

8. Find  $m\angle LTX$ .  
 $m\angle LTX = 7(6) = 42^\circ$

9. Find  $m\angle DLX$ .  
 $m\angle DLX = 180 - 96 = 84^\circ$

10. Find  $m\angle LTA$ .  
 $m\angle LTA = 180 - 42 = 138^\circ$